

Density of States and Cumulative Number of States

Before evaluating thermodynamic quantities (such as specific heat, internal energy, or chemical potential), we must establish the concept of the **Density of States (DOS)**, which serves as the fundamental bridge converting discrete quantum mechanical energy spectra into macroscopic thermodynamic integrals.

1. Formal Definition: Density of States $g(E)$

Consider a quantum system governed by a time-independent Hamiltonian $\hat{H}$ with a set of energy eigenvalues $\{E_n\}$ ($n = 0, 1, 2, \dots$).

The **Density of States** $g(E)$ (also written as $D(E)$) represents the number of quantum states per unit energy interval around an energy E . Formally, it is expressed as a sum of Dirac delta functions over all quantum eigenstates:

$$g(E) \equiv \sum_n \delta(E - E_n) \quad (1)$$

If an energy level E_n has a degeneracy d_n , the sum includes d_n identical delta terms.

Fundamental Property

The density of states allows discrete sums over quantum states to be converted into continuous integrals over energy space for any physical observable $f(E)$:

$$\sum_n f(E_n) = \int_0^\infty f(E) g(E) dE \quad (2)$$

2. Formal Definition: Cumulative Number of States $N(E)$

The **cumulative number of states** $N(E)$ (also known as the spectral staircase function or integrated density of states) counts the total number of single-particle quantum states with energies less than or equal to E :

$$N(E) \equiv \sum_n \Theta(E - E_n) \quad (3)$$

where $\Theta(x)$ is the Heaviside step function:

$$\Theta(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad (4)$$

$N(E)$ is a monotonically non-decreasing staircase function that jumps by 1 (or by the degeneracy d_n) at each discrete eigenvalue $E = E_n$.

Fundamental Relationship

The density of states $g(E)$ is the distribution derivative of the cumulative number of states $N(E)$ with respect to energy E :

$$g(E) = \frac{dN(E)}{dE} \quad (5)$$

Conversely, $N(E)$ is the running integral of $g(E)$:

$$N(E) = \int_0^E g(E') dE' \quad (6)$$

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3. Connection to Canonical Partition Function & Complex Integral Representation

In canonical statistical mechanics at inverse temperature $\beta = \frac{1}{k_B T}$, the single-particle quantum partition function $Z(\beta)$ is defined as:

$$Z(\beta) = \sum_n e^{-\beta E_n} \quad (7)$$

Substituting $g(E) = \sum_n \delta(E - E_n)$ transforms this discrete sum into a continuous integral:

$$Z(\beta) = \int_0^\infty g(E) e^{-\beta E} dE = \mathcal{L}\{g(E)\} \quad (8)$$

Thus, the canonical partition function $Z(\beta)$ is precisely the **Laplace transform** of the density of states $g(E)$ with respect to β .

Complex Contour Integral Representation (Bromwich Integral)

By applying the **Inverse Laplace Transform**, the exact density of states $g(E)$ can be extracted directly from the partition function $Z(\beta)$ via the **Bromwich complex contour integral**:

$$g(E) = \mathcal{L}^{-1}\{Z(\beta)\} = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} Z(\beta) e^{\beta E} d\beta \quad (9)$$

where γ is a real constant chosen to lie to the right of all singularities of $Z(\beta)$ in the complex β -plane.

Similarly, the cumulative number of states $N(E)$ is given by:

$$N(E) = \mathcal{L}^{-1}\left\{\frac{Z(\beta)}{\beta}\right\} = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} \frac{Z(\beta)}{\beta} e^{\beta E} d\beta \quad (10)$$

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4. Derivation of Smooth Density of States $\bar{g}(E)$ from Classical Partition Function

The exact quantum density of states $g(E)$ contains singular Dirac delta peaks at discrete energy levels. In macroscopic systems or high-temperature limits ($\hbar \rightarrow 0$), the discrete spectrum smoothes out into a continuous distribution $\bar{g}(E)$ (also written as $g_{\text{cl}}(E)$).

A. Extracting $\bar{g}(E)$ via Classical Partition Function $Z_{\text{cl}}(\beta)$

In classical statistical mechanics, a single quantum state occupies a fundamental phase-space volume of $(2\pi\hbar)^d = h^d$ in d dimensions. The classical canonical partition function $Z_{\text{cl}}(\beta)$ is:

$$Z_{\text{cl}}(\beta) = \frac{1}{(2\pi\hbar)^d} \iint e^{-\beta\mathcal{H}(\mathbf{r},\mathbf{p})} d^d r d^d p \quad (11)$$

Since $Z_{\text{cl}}(\beta)$ represents the smooth, high-temperature limit of $Z(\beta)$, taking its inverse Laplace transform yields the **smooth part of the density of states** $\bar{g}(E)$:

$$\bar{g}(E) = \mathcal{L}^{-1}\{Z_{\text{cl}}(\beta)\} = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} Z_{\text{cl}}(\beta) e^{\beta E} d\beta \quad (12)$$

Alternatively, inserting the phase-space integral for $Z_{\text{cl}}(\beta)$ directly gives the Thomas–Fermi formula:

$$\bar{g}(E) = \frac{1}{(2\pi\hbar)^d} \iint \delta(E - \mathcal{H}(\mathbf{r},\mathbf{p})) d^d r d^d p \quad (13)$$

B. Useful Inverse Laplace Transform Identities

To evaluate $\bar{g}(E)$ from partition functions $Z_{\text{cl}}(\beta)$, we use standard Laplace transform inversion formulas:

$$\mathcal{L}^{-1}\left\{\frac{1}{\beta^\alpha}\right\} = \frac{E^{\alpha-1}}{\Gamma(\alpha)} \quad (\text{for } \text{Re}(\alpha) > 0) \quad (14)$$

where $\Gamma(\alpha)$ is the Euler Gamma function.

1. Case $1/\beta^n$ (Integer Power $n \in \mathbb{Z}^+$): Since $\Gamma(n) = (n-1)!$:

$$\mathcal{L}^{-1}\left\{\frac{1}{\beta^n}\right\} = \frac{E^{n-1}}{(n-1)!} \quad (15)$$

2. Case $1/\beta^{n/2}$ (Half-Integer Power $n = 1, 3, 5, \dots$): Using $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ and $\Gamma(x+1) = x\Gamma(x)$:

- For $n = 1 \implies \alpha = 1/2$:

$$\mathcal{L}^{-1}\left\{\frac{1}{\beta^{1/2}}\right\} = \frac{E^{-1/2}}{\Gamma(1/2)} = \frac{1}{\sqrt{\pi E}} \quad (16)$$

- For $n = 3 \implies \alpha = 3/2$:

$$\mathcal{L}^{-1}\left\{\frac{1}{\beta^{3/2}}\right\} = \frac{E^{1/2}}{\Gamma(3/2)} = \frac{E^{1/2}}{\frac{1}{2}\sqrt{\pi}} = \frac{2\sqrt{E}}{\sqrt{\pi}} \quad (17)$$

- For $n = 5 \implies \alpha = 5/2$:

$$\mathcal{L}^{-1}\left\{\frac{1}{\beta^{5/2}}\right\} = \frac{E^{3/2}}{\Gamma(5/2)} = \frac{E^{3/2}}{\frac{3}{4}\sqrt{\pi}} = \frac{4E^{3/2}}{3\sqrt{\pi}} \quad (18)$$

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5. Applications

A. Density of States for a 3D Free Particle

The classical partition function for a free particle of mass m confined in a 3D volume V ($\mathcal{H} = \frac{p^2}{2m}$) is:

$$Z_{\text{cl}}(\beta) = \frac{V}{(2\pi\hbar)^3} \int_{-\infty}^{\infty} e^{-\frac{\beta p^2}{2m}} d^3p = V \left(\frac{m}{2\pi\hbar^2\beta} \right)^{3/2} \quad (19)$$

Rearranging to isolate β :

$$Z_{\text{cl}}(\beta) = V \left(\frac{m}{2\pi\hbar^2} \right)^{3/2} \beta^{-3/2} \quad (20)$$

Applying the inverse Laplace transform $\bar{g}(E) = \mathcal{L}^{-1}\{Z_{\text{cl}}(\beta)\}$:

$$\bar{g}(E) = V \left(\frac{m}{2\pi\hbar^2} \right)^{3/2} \mathcal{L}^{-1}\{\beta^{-3/2}\} \quad (21)$$

Using the identity $\mathcal{L}^{-1}\{\beta^{-3/2}\} = \frac{2\sqrt{E}}{\sqrt{\pi}}$:

$$\bar{g}(E) = V \left(\frac{m}{2\pi\hbar^2} \right)^{3/2} \left(\frac{2\sqrt{E}}{\sqrt{\pi}} \right) \quad (22)$$

Simplifying yields the standard smooth 3D free-particle density of states:

$$\bar{g}(E) = \frac{V}{4\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \sqrt{E} \quad (23)$$

B. Density of States for Harmonic Oscillators

1. 1D Harmonic Oscillator: The classical partition function for a 1D harmonic oscillator with frequency ω is:

$$Z_{\text{cl, 1D}}(\beta) = \frac{1}{\beta\hbar\omega} \quad (24)$$

Applying the inverse Laplace transform using $\mathcal{L}^{-1}\{\beta^{-1}\} = 1$:

$$\bar{g}_{1D}(E) = \mathcal{L}^{-1}\left\{ \frac{1}{\hbar\omega} \beta^{-1} \right\} = \frac{1}{\hbar\omega} \quad (25)$$

Result: The smooth density of states for a 1D harmonic oscillator is constant, independent of energy E .

2. 3D Isotropic Harmonic Oscillator: The classical partition function for a 3D isotropic harmonic oscillator is:

$$Z_{\text{cl, 3D}}(\beta) = (Z_{\text{cl, 1D}}(\beta))^3 = \frac{1}{(\hbar\omega)^3\beta^3} \quad (26)$$

Applying the inverse Laplace transform using the identity $\mathcal{L}^{-1}\{\beta^{-3}\} = \frac{E^2}{2!}$:

$$\bar{g}_{3D}(E) = \frac{1}{(\hbar\omega)^3} \mathcal{L}^{-1}\{\beta^{-3}\} = \frac{1}{(\hbar\omega)^3} \frac{E^2}{2} \quad (27)$$

$$\bar{g}_{3D}(E) = \frac{E^2}{2(\hbar\omega)^3} \quad (28)$$

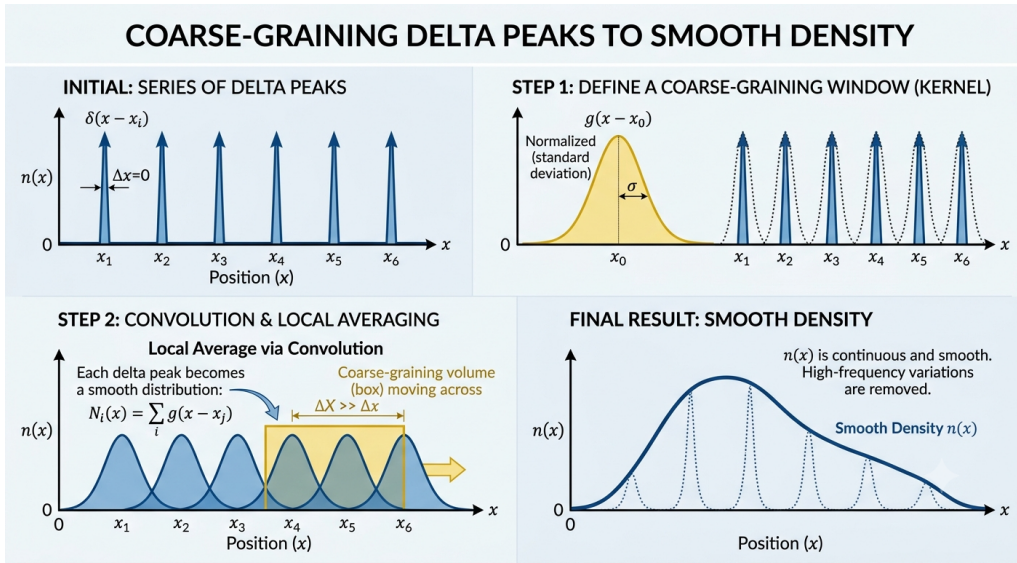


Figure 1: Schematic representation of the quantum density of states. The exact spectrum $g(E)$ consists of sharp Dirac delta peaks at eigenenergies E_n . The smooth classical density of states $\bar{g}(E)$, extracted via the inverse Laplace transform of $Z_{cl}(\beta)$, acts as a continuous envelope averaging through the discrete quantum spectrum.